Name: _____ Matr.-Nr.: _____

Microwave Engineering WiSe 2022/23

Important general information:

1. Enter your name and your matriculation number **on this and on each following sheet** with answers to the exam questions.
2. Do not use red color or lead pencils.
3. Place your student identity card clearly visible on the table.
4. You have **120 minutes** to complete the tasks.
5. Before you start, check that your copy of the exam contains **x pages** and the paper **Evolution of the Nonuniform Distributed Power Amplifier**.
6. For each value you take from the data sheet, clearly mark them on the solution sheet ("Value from page X of data sheet") and on the data sheet (circle the value and write "Value for task X.Y")
7. Further solution sheets are available on request.

Teil	1 (Paper)	2 (General)		Sum
Task	1	2	3	
Maximum Credits	x	x	x	100
Achieved Credits				

Grade: _____

Name: _____ Matr.-Nr.: _____

Part 1: Paper

Charles F. Champbell, "Evolution of the Nonuniform Distributed Power Amplifier" IEEE Microwave Magazine, 2019

Task 1 – 39.5 Credits

- **1.1** What is the advantage of Gallium Nitride (GaN) transistors? (2 credits)
  increase in output power 1 and power handling 1
- **1.2** Why is silicon or silicon carbide for the epitaxial growth of the transistors especially beneficial for power amplifiers (PAs)? (1 credit)
  because they are high-thermal conductivity substrates and therefore beneficial for high power densities 11
- **1.3** What is the problem of high-voltage operation of GaN transistors with respect to wide-band design? (1 credits)
  The high -voltage operation generates a high-Q equivalent circuit and therefore the achievable bandwidth is limited according to the Bode-Fano criteria
- **1.4** What combination was the reason for the frequent use of nonuniform distributed amplifier topology for wide-band GaN PAs? (2 credits)
  high-output power density 1 and the Bode-Fano bandwidth limitations associated with reactive matching 1
- **1.5** Please draw the Norton-equivalent circuit of a microwave power device. The input can be considered ideal. (3 credits)
  ideal VCCS 1, Rp 1, Cp 1
- **1.6** How can the output conductance and output capacitance of the GaN power device be measured? (1 credits)
  load pull
- **1.7** Assuming we have a GaN transistor with a maximum current capability of 5 A, a knee voltage of 100 mV and we apply a drain supply of 30 V. The parallel output capacitance of the Norton equivalent circuit has a value of 2 pF and the fitting parameter equals 1.5. What 20 dB matching bandwidth can we achieve according to the paper? (6.5 credits)
  BW< (4.343 Imax)/(RL(dB) Cp β (VDsupply-Vknee))= (4.343x 5 A)/(20x200 fF x 1.5 x (30 V-100 mV))=12.1 GHz
  0.5 for each correct value in equation, -0.25 for wrong unit, 3 for correct result with unit
- **1.8** What are options to increase the theoretically achievable bandwidth if we want to use the GaN transistor from question 1.7 in a classical common-source amplifier configuration? (2 credits)
  lower supply voltage 1, accept a worse reflection coefficient 1
- **1.9** What is the fundamental idea behind distributed amplification in order to increase the Bode-Fano limitation? (1 credit)
  reduce Cp, i.e. compensate for the in- and output capacitance of the transistors 1
- **1.11** What are the crucial components to achieve the increase in the Bode-Fano limitation for a distributed amplifier? (1 credit)
  artificial transmission lines 1

Name: _____ Matr.-Nr.: _____

- **1.12** The maximum current of the GaN device from 1.7 is valid for 5 µm total gate width (TGW). Normalized to the device size, we have a current capability of this process of 5 A/ 5 µm = 1000 A / mm. What is the real part of the Norton equivalent circuit for the transistor from question 1.7? (6 credits)
  Rp=β(VDsupply-Vk)/(Imax(A/mm))=1.5(30 V-0.1 V)/ 1000 A/mm=0.04485 Ωmm
  0.5 for each correct value in equation, -0.25 for wrong unit, 3 for correct result with unit
- **1.13** Is usually the input or output capacitance of a transistor higher? What does this mean for building a distributed amplifier and how can this this problem be solved? (3 credits)
  input 1; gate line is bandwidth limiting 1, use additional series capacitances at the gate 1
- **1.14** What are three reasons that uniform distributed amplifiers perform well as low noise amplifiers (LNAs), but have a reduced performance as PAs? (1 credits)
  dissipation in the drain line termination 1, RF signal decreases moving down the gate line 1, transistor loading will vary from device to device and over frequency 1
- **1.15** What circuit topology adapts a distributed amplifier topology the PA needs? What is required for the series gate capacitance and the characteristic impedance of the output transmission line? With which geometrical parameter can the characteristic impedance be influenced in the layout? (6 credits)
  nonuniform distributed power amplifier 2taper the capacitances and transmission lines 2, i.e. smaller capacitance near the input, highest characteristic impedance near input, line width affects the characteristic impedance 2
- **1.16** Which two transformer configurations are discussed in the paper with respect to bandwidth limitation and which additional option is used in the design shown in the figure 13 of the paper? (3 credits)
  Ruthroff 1 and Trifilar 1 transformer, lambda quarter transformer 1

Name: _____ Matr.-Nr.: _____

Part 2: General

Task 2 – 20 Credits

In Figure 1, the measurements of a PA are shown.

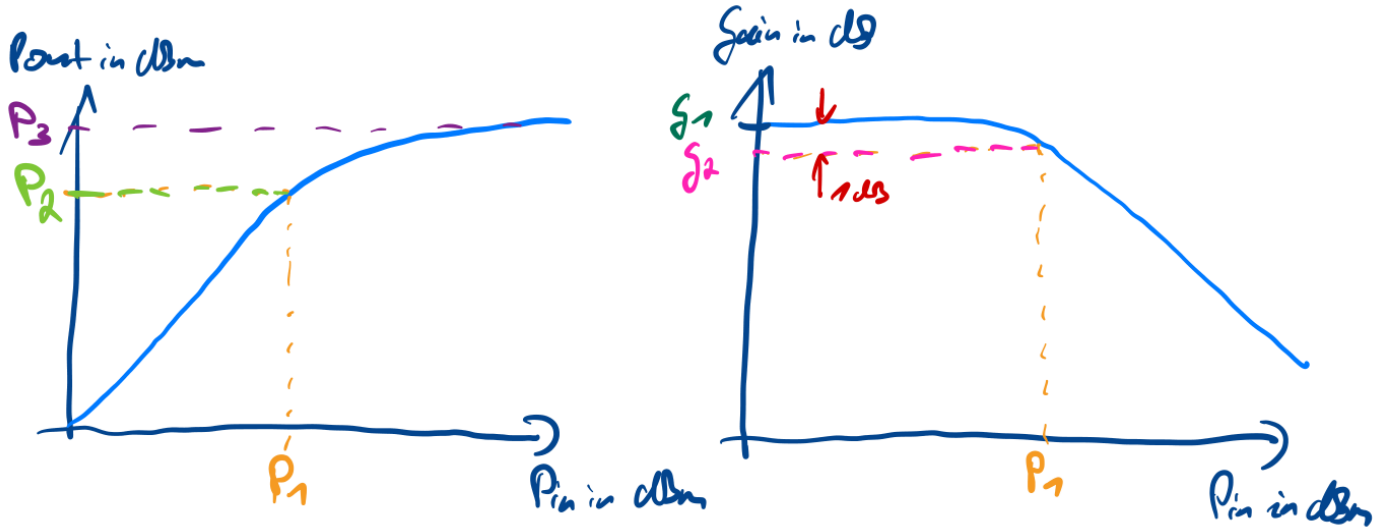


Figure 1. Large-signal measurements of a PA.

- **2.1** Give a generic signal, which can be applied at the PAs input to generate such plots. How is this measurement signal usually referred to? (3 credits)
  uin=A cos(wt) 2 (1 amplitude, 1 cos function), one-tone signal
- **2.2** Which metrics of a PA relate to indicated power and gain points P_1, P_2, P_3 and G_1? (5 credits)
  P1 IP1dB, P2 OP1dB, P3 Psat, G1small signal gain, G2 gain at P1dB; each 1
- **2.3** Give the equation, how to calculate P_2 from P_1 and G_2. (2 credits)
  P2=P1+G2 P1 and G2 each 1
- **2.4** Which of the indicated power points P_2 from P_3 are more relevant for communication amplifiers and for radar amplifiers? (2 credits)
  P1dB communication, Psat radar; each 1
- **2.5** What is required to do with P_{in} in order to measure the linear behavior of a PA? Why is it possible to force a linear operation by this measure? (2 credits)
  use a very low input power 1, because nonlinear effects are drive level dependent 1
- **2.6** What do you need to apply at the PAs input, in order to generate intermodulation (IM) products? (2 credits)
  apply two tones at the input, i.e. to closely spaced frequencies 1, which drive the PA in the nonlinear regime 1
- **2.7** What is the most harmful IM product for in-band amplification and why is it the most harmful? Which additional large-signal metric of an amplifier can you derive from this IM products? (3 credits)
  IM3 product 1, close to the carrier 1, OIP3 1
- **2.7** Are S-parameter measurements small- or large-signal metrics? (1 credit)
  small signal

Name: _____ Matr.-Nr.: _____

Task 3 – 24 Credits

Figure 2 shows the transformation path of a 2-element matching for a given frequency which transforms match Z_{out} into Z_{goal}.

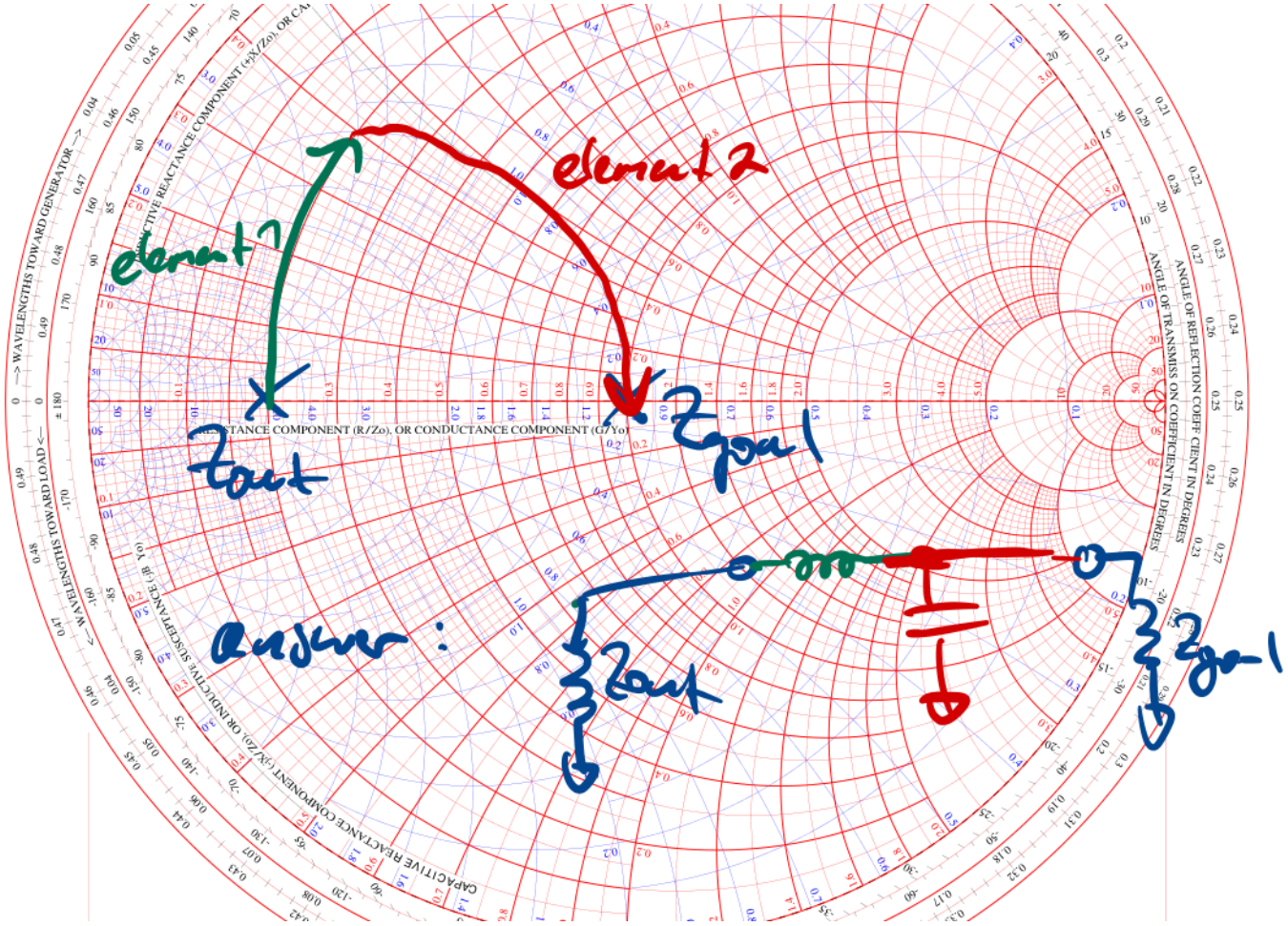


Figure 2. Transformation path of a 2-element matching.

- **3.1** Draw the matching network with LC-elements including the terminations. Which of the element is in shunt and which element is in series configuration? (8 credits)
  each freq dependent element 2 and term each 1in the schematic, Cp is shunt 1, Lser is series 1
- **3.2** At millimeter-wave frequencies, so-called stubs are frequently used to build matching networks. Which matching element from Figure 2 can you replaced by a stub and which kind of stub do you need to use? (2 credits)
  Open stub 1 can replace shunt capacitance 1

Name: _____ Matr.-Nr.: _____

We want to investigate another transformation path to match Z_{out} to Z_{goal}. Now the schematic of the matching network is given in Figure 3.

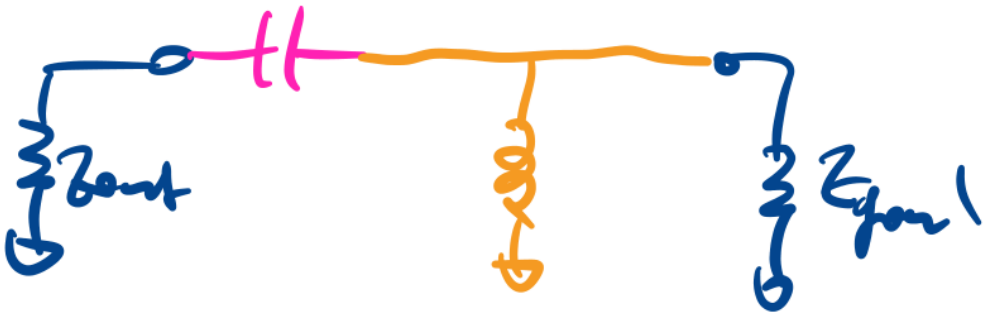
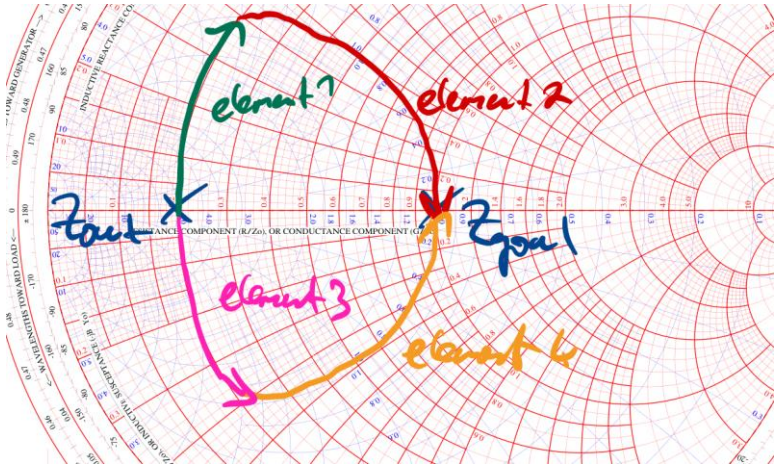


Figure 3. Schematic of an alternative matching network to transform Z_{out} to Z_{goal}.

- **3.3** Draw the transformation path of the LC matching from Figure 3 into the Smith-chart. (4 credits)

answer: 2 for each



- **3.4** What can we expect for the matching bandwidth of both networks? Which criterion is relevant to determine this and how can this criterion be graphically represented? (6 credits)
  both have the same bandwidth 2, Bode-Fano 2, constant quality factors in smith chart 2
- **3.5** How can you transform Z_{out} to Z_{goal} with only one matching element? (2 credits)
  with a lambda-quarter transformer 2
- **3.6** What replaces our inductances at millimeter-wave frequencies? (1 credits)
  transmission liness

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